Details About Limit and Functions

Key Points

Introduction to Limit and Functions

Details about Limit

- **Types of limit**
- **Limit Laws**
- **L'Hopital's Rule**

Details about Functions

Relation Between Limit and Functions

Real-Life Applications of Limit and Functions

Conclusion and Q&A

Introduction to Limit and Functions

- Limit : A limit describes the value a function approaches as the input approaches a certain point.

  Example: $\lim_{x \to a} f(x) = L$

- Function : Function is an expression between an independent variable and dependent variable .

  Example : $f(x) = y$

Types of Limit

There are several types of limit. But , today we will discuss about two types of limit . These are One-sided Limit and Two-sided Limit.

- One-sided Limit
  - Left-hand limit:

    For Left-hand limit, we approach a point from the left.

    Example $\lim_{x \to 2-} f(x)$
  - Right-hand limit:

    For Right-hand limit, we approach a point from the right.

    Example $\lim_{n \to 2+} f(x)$
- Two-sided Limit:

  For a two-sided limit, we approach from both directions.

  Example $\lim_{n \to 2+} f(x)$ (from the right) **and** $\lim_{x \to 2-} f(x)$ (from the left)

Limit Laws

- Sum Law: The limit of a sum of functions equals the sum of their limits.
  Example : $\lim_{x \to c}[f(x) + g(x)] = \lim_{x \to c} f(x) + \lim_{x \to c} g(x)$

- Difference Law: The limit of a Difference of functions equals the Difference of their limits.
  Example: $\lim_{x \to c}[f(x) - g(x)] = \lim_{x \to c} f(x) - \lim_{x \to c} g(x)$

- Product Law: The limit of a Product of functions equals the Product of their limits.
  Example: $\lim_{x \to c}[f(x) \cdot g(x)] = \lim_{x \to c} f(x) \cdot \lim_{x \to c} g(x)$

Limit Laws

- Division Law: The limit of a Product of functions equals the Product of their limits, provided the denominator's limit is not zero .
  Example: $\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{\lim_{x\to c} f(x)}{\lim_{x\to c} g(x)}, \quad \text{if} \lim_{x\to c} g(x) \neq 0$

L'Hospital's Rule

If f(x) and g(x) be two continuous functions at x=a and also their derivatives f'(x) and g'(x) be such that (f(a)=g(a)=0 $\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)}$ provided g'(a) is not equal to 0 .

It states that if: $\lim_{x\to a} f(x)=\lim_{x\to a} g(x)=0 \text{ or } \infty$ and both functions f(x) and g(x) are differentiable near a, then

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

L'Hospital's Rule

Let’s take an example: $\lim\limits_{x \to 0} \frac{\sin(x)}{x}$

Direct substitution gives 0/0, an indeterminate form.

Applying L'Hôspital's Rule:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = \lim_{x \to 0} \frac{\cos(x)}{1} = \cos(0) = 1$$

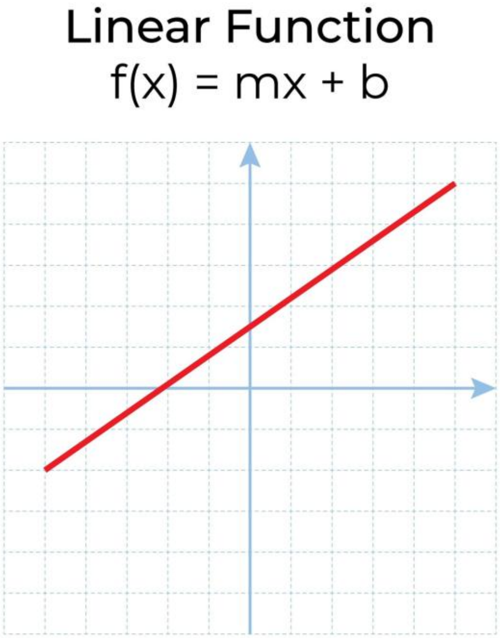
This powerful theorem helps resolve many such cases.

Types of Functions

- Linear Function:

A linear function is a function whose graph is a line

Example : $f(x) = mx + b$.

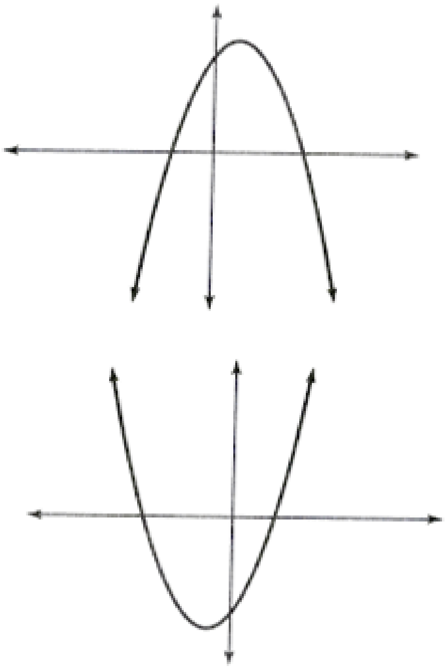


Types of Functions

- Quadratic Function: A quadratic function is defined as a polynomial where the highest degree of any variable is 2

Example:

$f(x) = ax^2 + bx + c.$

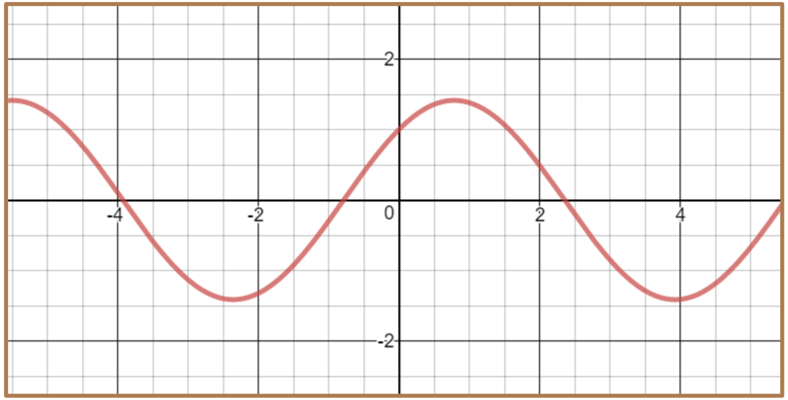


Types of Functions

• Trigonometric Function:

Trigonometric functions are those functions which contain trigonometric properties like sin, cos, tan, cot, sec, and cosec.

Example: f(x)=sin x



Domain, Co-domain and Range

Domain is the set of all possible inputs of a function . Co-domain is the set of all possible outputs of a function . And range is the set of the actual outputs of a function.

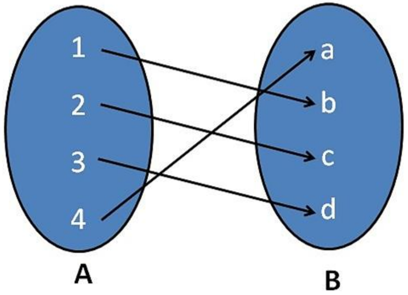
Example (Domain and Range):

For f(x)= $\frac{1}{x}$ the domain will be {$\mathbb{R}$-0} and range will be {$\mathbb{R}$-0} .

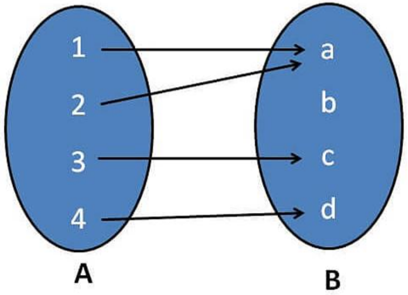
One One and Onto Functions

One One:

A function will be one one if no two different input values map to the same output value. Otherwise it will not be one one .



One One

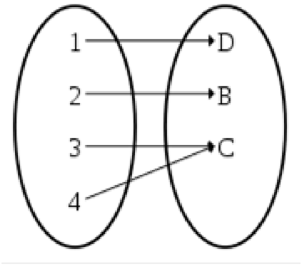


Not One One

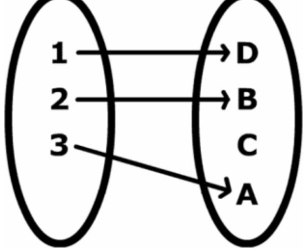
One One and Onto Functions

Onto:

A function will be onto if every element in the codomain has at least one corresponding element in the domain. Otherwise it will not be onto .



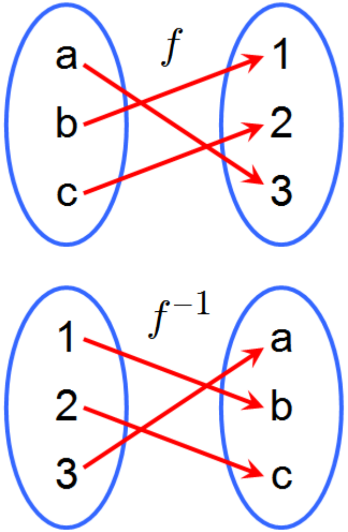
Onto



Not Onto

Inverse of Functions

Let, $f:A \longrightarrow B$ be a function which is one-one and onto . Then inverse function of $f:A \longrightarrow B$ be denoted as $f^{-1}:B \longrightarrow A$



Continuity of Functions

A function f(x) is continuous at x=a if:

Left Hand Limit of f(x) = Right Hand Limit of f(x) = f(a)

Or,

$\lim_{x\to a-} f(x) = \lim_{x\to a+} f(x) = f(a)$

Relationship Between Limits and Functions with Real-Life Application

On this slide, I’ll briefly explain the relationship between limits and functions and their real-life application. A function represents a relationship between input and output, while a limit helps us understand how the function behaves as the input approaches a specific value.

For example, $$\lim_{x \to 0} \frac{1}{x}$$

Even though the function doesn't exist at 0, the limit shows that the function approaches infinity. In real life, limits are crucial in physics. They help calculate instantaneous velocity, like determining how fast a car is moving at a specific point in time. This would be impossible to measure without the concept of limits."

Any Questions ?

Thank You.